

PREPARED FOR

Ashesi University — Classroom Booking System Project
Contact: Project Stakeholder / Booking System Lead

Project Summary

Comprehensive delivery covering production bug fixes, feature enhancements, UI/UX polish, architecture documentation, AI-assisted engineering, human architecture review, CI/CD pipeline, cloud infrastructure, security, privacy compliance, and quality assurance for the AURA platform.

1. AI-Assisted Engineering & Tooling

Advanced AI agents and IDE assistants used to accelerate development, review, and documentation.

Tool / service	Rate (GHS)	Hours / Units	Amount (GHS)
Kimi Code CLI — primary agentic development, debugging, commits	500	35 hrs	17,500
Anthropic Claude — code review, architecture reasoning, documentation polish	500	12 hrs	6,000
OpenAI Codex / GitHub Copilot — code completion, test scaffolding, refactor suggestions	500	8 hrs	4,000
Subtotal — AI-assisted engineering			27,500

2. Human Architecture Design & Review

Senior engineering oversight, system design, cross-module review, and production-readiness sign-off.

Activity	Rate (GHS)	Hours	Amount (GHS)
System architecture review & bounded-context validation	800	6	4,800
Code review (availability engine, booking race conditions, UI fixes)	800	5	4,000
Production-readiness & deployment planning	800	3	2,400
Subtotal — human architecture & review			11,200

3. Feature Development & Bug Fixes

Work package	Rate (GHS)	Hours	Amount (GHS)
Calendar dayBoundaries fix (Schedule-X whole-hour validation, grid tuning)	500	6	3,000
Booking dialog enhancement (editable room/date/time, availability pre-check)	500	22	11,000
UI/UX fixes (uniform button radius, overlay z-index for popovers/selects/tooltips)	500	8	4,000
Testing, QA & typecheck (Vitest, TypeScript, integration validation)	500	9	4,500

Work package	Rate (GHS)	Hours	Amount (GHS)
Subtotal — feature development			22,500

4. DevOps, CI/CD & Cloud Infrastructure

Service / activity	Rate (GHS)	Hours / Units	Amount (GHS)
GitHub — repository management, PR review workflow, branch protection	500	4 hrs	2,000
GitHub Actions CI/CD — build, test, lint, sqlc generate checks	500	6 hrs	3,000
Vercel — web app deployment, preview environments, edge config	500	3 hrs	1,500
Render — Go API + PostgreSQL Blueprint deployment, environment wiring	500	4 hrs	2,000
PostgreSQL — managed DB setup, migration, replica routing	500	3 hrs	1,500
Subtotal — DevOps & infrastructure			10,000

5. Documentation & Release Management

Deliverable	Amount (GHS)
Architecture documentation update (ARCHITECTURE.md core-algorithms section)	2,500
PDF generation (ARCHITECTURE.pdf, billing quote)	1,500
Git commits, release notes, change tracking	1,000
Subtotal — documentation & release	5,000

6. Security, Privacy & Compliance

Service	Amount (GHS)
Security review (RBAC, JWT, CSRF, rate limiting, audit logging)	4,000
Privacy & Ghana Data Protection Act 843 alignment review	2,500
Booking double-spend / race-condition guarantee validation	2,000
Subtotal — security & privacy	8,500
Total before tax	84,700
VAT / NHIL / GETFund (estimated 22.5%)	19,057.50
GRAND TOTAL	GHS 103,757.50

7. Technologies, Platforms & Languages

Kimi Code CLI

Claude

Codex / Copilot

GitHub

GitHub Actions

Next.js 16

React 19

TypeScript

Go

PostgreSQL 18

Tailwind CSS v4

Radix UI

Schedule-X

React Hook Form

TanStack Query

sqlc

Vitest

Playwright

Vercel

Render

Pandoc

WeasyPrint

Platforms delivered

- Web requester portal (Next.js App Router)
- Admin console (RBAC-gated)
- Mobile app (Expo SDK 56 / NativeWind)
- REST API (Go / chi router)
- Cloud deploy: Vercel + Render Blueprint

Languages & data

- TypeScript / JavaScript (web + mobile)
- Go (backend services)
- SQL (PostgreSQL migrations & queries)
- CSS / Tailwind (design system)
- OpenAPI 3.1 (contract-first clients)

8. Features Available

- Room search & availability grid
- Booking request & approval workflow
- Unified calendar (day/week/month)
- Maintenance windows
- Timetable CSV/XLSX import
- Notifications (SSE + email + push)
- Utilisation & conflict reports
- Admin override & audit log
- User/department/role management
- Semester & room catalogue CRUD

Quality gates run: TypeScript type-checking, Go vet, Vitest unit tests (74 passed), integration validation, manual cross-browser checks, AI-assisted review, human architecture sign-off, and production-deploy readiness review.

Quote prepared by the AURA engineering team · 7 July 2026 · All figures in Ghana Cedis (GHS).
Payment terms: 50% on acceptance, 50% on final delivery. Bank details provided on invoice.